Course :- Discrete Mathematics and Graph Theory :-

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Problem ① :-

$A = \{$All positive divisors of 18$\}$

$B = \{$All positive multiples of 4 less than 25$\}$

$C = \{$All positive common divisors of 24 and 36$\}$

$A = \{1, 2, 3, 6, 9, 18\}$

$B = \{4, 8, 12, 16, 20, 24\}$

$C = \{1, 2, 3, 4, 6, 12\}$

PROVE $A \Delta (B \Delta C) = (A \Delta B) \Delta C$.

Step ① :- $B \Delta C = (B - C) \cup (C - B)$

$= \{8, 16, 20, 24\} \cup \{1, 2, 3, 6\}$

$= \{1, 2, 3, 6, 8, 16, 20, 24\}$

Step ② :- $A \Delta (B \Delta C) = A \Delta (X)$ $\{ : X = \{1, 2, 3, 6, 8, 16, 20, 24\}$

$= (A - X) \cup (X - A)$

$= \{9, 18\} \cup \{8, 16, 20, 24\}$

$= \{8, 9, 16, 18, 20, 24\}$.

Step ③ :- $A \Delta B = (A - B) \cup (B - A)$

$= \{1, 2, 3, 4, 6, 9, 9, 12, 16, 18, 20, 24\}$ $[\because A \cap B = \emptyset]$

Step ④ :- $(A \Delta B) \Delta C$

$= (Y - C) \cup (C - Y)$ $(\because Y = A \Delta B)$

$= \{8, 9, 16, 18, 20, 24\}$

Step ⑤ :- Conclusion of Step ② = Step ④

$A \Delta (B \Delta C) = (A \Delta B) \Delta C$

Problem ②

$S = \{$ Ram, Shyam, Ramu, Ajit, Roy $\}$

$R_1 = \{(x, y) : x$ and y have at least two letters in common$\}$

$R_1 = \{$(Ram, Ram), (Ram, Shyam), (Ram, Ramu), (Shyam, Ram), (Shyam, Shyam), (Shyam, Ramu), (Ramu, Ramu), (Ramu, Shyam), (Ramu, Ram), (Ajit, Ajit), (Roy, Roy)$\}$

Properties R_1 :- Reflexive = yes

Symmetric = yes

Transitive = yes

Anti-symmetric = No

Asymmetric = No

Irreflexive = No.

Adjacency Matrix :-

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$R_2 = \{(x, y) : x$ and y have same length$\}$

Ram = 3, Shyam = 5, Ramu = 4, Ajit = 4, Roy = 3.

Ram & Roy Ramu & Ajit ? Each element with itself :

Reflexive = yes Transitive = yes Asymmetric = No

Symmetric = yes Antisymmetric = No Irreflex = No.

Adjacency Matrix :- $M(R_2) = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}$

$R_3 = \{(a, b) :$ a and b have a vocal in common$\}$

- Ram, Shyam, Ramu, Ajit all share vocal "a" in common.
- Roy has only vocal "o".

Properties : Reflexive : yes, Symmetric : yes, Transitive : yes.

Antisymmetric : No, Assymmetric : No

Adjacency Matrix :- $\begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

$\boxed{\text{Problem ③}}$ $A_1 = \{1,2,3,4\}$ $A_2 = \{4,5,6\}$ $A_3 = \{6,7,8\}$

$$R_1 = \{(x,y) \in A_1 \times A_2 : y - x = 2\}$$

$$R_2 = \{(x,y) \in A_2 \times A_3 : y - x = 2\}$$

$$R_1 = \{(2,4), (3,5), (4,6)\}$$

$$R_2 = \{(4,6), (5,7), (6,8)\}$$

<u>Adjacency matrix of R_1</u> :-

$$\begin{array}{c|ccc} & 4 & 5 & 6 \\ \hline 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 3 & 0 & 1 & 0 \\ 4 & 0 & 0 & 1 \end{array}$$

<u>Adjacency matrix of R_2</u> :-

$$\begin{array}{c|ccc} & 6 & 7 & 8 \\ \hline 4 & 1 & 0 & 0 \\ 5 & 0 & 1 & 0 \\ 6 & 0 & 0 & 1 \end{array}$$

Based on definition of composition (x,z) if $(x,y)\,(y,z) \in R_1$ & R_2

$$\therefore \quad R_1 \circ R_2 = \{(2,6), (3,7), (4,8)\}.$$

Problem ④ :-

$$\begin{array}{c} (a, b, c, d, e) \\ \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix} \end{array} \qquad \begin{array}{l} a \to a \\ b \to b \\ c \to a, b, c, d \\ d \to a, b, d \\ e \to a, b, c, d, e \end{array}$$

a, b are smallest
d is above a & b
c is " a, b, d
e > all.

Showing immediate relations :-

$a < d$
$b < d$
$d < c$
$c < e$

Problem ⑤ :-

To prove $(P(S), \leq)$ is a partial order, we must show that the relation $\leq$ defined by $\leq = \{(S_1, S_2) : S_1 \subseteq S_2\}$ on $P(S)$ satisfies these below properties:

→ reflexive
→ antisymmetric
→ transitive.

Proof :- ① Let $A, B, C \in P(S)$, (A, B, C are all subsets)

For every $A \in P(S)$, $A \subseteq A$

Hence $(A, A) \in \leq$.

$\therefore$ $\leq$ is reflexive.

② Suppose $A \leq B$ and $B \leq A$. then by definition,

$A \subseteq B$ and $B \subseteq A$, so $A = B$

$\therefore$ $\leq$ is assymmetric.